

3.5.1 Newton's laws of motion

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) Newton's three laws of motion	HSW7
(b) linear momentum; $p = mv$; vector nature of momentum	
(c) net force = rate of change of momentum; $F = \frac{\Delta p}{\Delta t}$	Learners are expected to know that $F = ma$ is a special case of this equation. HSW9, 10 M2.1, M3.9
(d) impulse of a force; impulse = $F\Delta t$	
(e) impulse is equal to the area under a force–time graph.	Learners will also be expected to estimate the area under non-linear graphs. HSW3 Using a spreadsheet to determine impulse from F – t graph. M3.8, M4.3